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1  clear; clc;
2
3  dragscalingfactor = 12.52;
4  liftscalingfactor = 89.49;
5  dragscalingfactor2 = 12.47;
6  liftscalingfactor2 = 72.48;
7
8  toolUncertainty = 0.01;    % N
9
10 data150    = readmatrix('150deg.xlsx');
11 data165    = readmatrix('165deg.xlsx');
12 data180    = readmatrix('180deg.xlsx');
13 datastand  = readmatrix('teststand1data.xlsx');
14 datastand2 = readmatrix('teststand2data.xlsx');
15
16 area150 = 5.02682e-3;
17 area165 = 5.28867e-3;
18 area180 = 5.37630e-3;
19
20 n = 10;
21 t95 = 2.262;
22
23 vel150     = data150(:,2);
24 drag150     = data150(:,3);
25 dragstd150 = data150(:,4);
26 lift150     = data150(:,5);
27 liftstd150 = data150(:,6);
28
29 vel165     = data165(:,2);
30 drag165     = data165(:,3);
31 dragstd165 = data165(:,4);
32 lift165     = data165(:,5);
33 liftstd165 = data165(:,6);
34
35 vel180     = data180(:,2);
36 drag180     = data180(:,3);
37 dragstd180 = data180(:,4);
38 lift180     = data180(:,5);
39 liftstd180 = data180(:,6);
40
41 velstand    = datastand(:,2);
42 dragstand    = datastand(:,3);
43 dragstdstand = datastand(:,4);
44 liftstand    = datastand(:,5);
45 liftstdstand = datastand(:,6);
46
47 velstand2    = datastand2(:,2);
48 dragstand2    = datastand2(:,3);
49 dragstdstand2 = datastand2(:,4);
50 liftstand2    = datastand2(:,5);
51 liftstdstand2 = datastand2(:,6);
52
53 % Force calculations
54 forcedrag165 = zeros(length(drag165),1);
55 forcelif165  = zeros(length(lift165),1);
56 forcedrag180 = zeros(length(drag180),1);
57 forcelif180  = zeros(length(lift180),1);
58 forcedrag150 = zeros(length(drag150),1);
59 forcelif150  = zeros(length(lift150),1);
60
61 for i = 1:length(vel165)

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62     forcedrag165(i) = dragscalingfactor * ((drag165(i) - drag165(1)) - (dragstand(i) -
dragstand(1)));
63     forcelif165(i) = liftscalingfactor * ((lift165(i) - lift165(1)) - (liftstand(i) -
liftstand(1)));
64 end
65
66 for i = 1:length(vel180)
67     forcedrag180(i) = dragscalingfactor * ((drag180(i) - drag180(1)) - (dragstand(i) -
dragstand(1)));
68     forcelif180(i) = liftscalingfactor * ((lift180(i) - lift180(1)) - (liftstand(i) -
liftstand(1)));
69 end
70
71 for i = 1:length(vel150)
72     forcedrag150(i) = dragscalingfactor2 * ((drag150(i) - drag150(1)) - (dragstand2(i) -
dragstand2(1)));
73     forcelif150(i) = liftscalingfactor2 * ((lift150(i) - lift150(1)) - (liftstand2(i) -
liftstand2(1)));
74 end
75
76 % 95% confidence interval in FORCE
77 dragCIforce150 = zeros(length(drag150),1);
78 liftCIforce150 = zeros(length(lift150),1);
79 dragCIforce165 = zeros(length(drag165),1);
80 liftCIforce165 = zeros(length(lift165),1);
81 dragCIforce180 = zeros(length(drag180),1);
82 liftCIforce180 = zeros(length(lift180),1);
83
84 for i = 1:length(vel150)
85     se_drag_150 = sqrt((dragstd150(i)^2 + dragstd150(1)^2 + dragstdstand2(i)^2 +
dragstdstand2(1)^2) / n);
86     se_lift_150 = sqrt((liftstd150(i)^2 + liftstd150(1)^2 + liftstdstand2(i)^2 +
liftstdstand2(1)^2) / n);
87     dragCIforce150(i) = sqrt((t95 * dragscalingfactor2 * se_drag_150)^2 + toolUncertainty^2);
88     liftCIforce150(i) = sqrt((t95 * liftscalingfactor2 * se_lift_150)^2 + toolUncertainty^2);
89 end
90
91 for i = 1:length(vel165)
92     se_drag_165 = sqrt((dragstd165(i)^2 + dragstd165(1)^2 + dragstdstand(i)^2 +
dragstdstand(1)^2) / n);
93     se_lift_165 = sqrt((liftstd165(i)^2 + liftstd165(1)^2 + liftstdstand(i)^2 +
liftstdstand(1)^2) / n);
94     dragCIforce165(i) = sqrt((t95 * dragscalingfactor * se_drag_165)^2 + toolUncertainty^2);
95     liftCIforce165(i) = sqrt((t95 * liftscalingfactor * se_lift_165)^2 + toolUncertainty^2);
96 end
97
98 for i = 1:length(vel180)
99     se_drag_180 = sqrt((dragstd180(i)^2 + dragstd180(1)^2 + dragstdstand(i)^2 +
dragstdstand(1)^2) / n);
100    se_lift_180 = sqrt((liftstd180(i)^2 + liftstd180(1)^2 + liftstdstand(i)^2 +
liftstdstand(1)^2) / n);
101    dragCIforce180(i) = sqrt((t95 * dragscalingfactor * se_drag_180)^2 + toolUncertainty^2);
102    liftCIforce180(i) = sqrt((t95 * liftscalingfactor * se_lift_180)^2 + toolUncertainty^2);
103 end
104
105 % Coefficients
106 cd150 = zeros(length(forcedrag150),1);
107 cl150 = zeros(length(forcelif150),1);
108 cd165 = zeros(length(forcedrag165),1);
109 cl165 = zeros(length(forcelif165),1);
110 cd180 = zeros(length(forcedrag180),1);

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111 cl180 = zeros(length(forcelift180),1);
112
113 cdCI150 = zeros(length(forcedrag150),1);
114 clCI150 = zeros(length(forcelift150),1);
115 cdCI165 = zeros(length(forcedrag165),1);
116 clCI165 = zeros(length(forcelift165),1);
117 cdCI180 = zeros(length(forcedrag180),1);
118 clCI180 = zeros(length(forcelift180),1);
119
120 p1 = 1.184;
121 p2 = 1.176;
122
123 for i = 1:length(forcedrag150)
124     if vel150(i) == 0
125         cd150(i) = NaN;
126         cl150(i) = NaN;
127         cdCI150(i) = NaN;
128         clCI150(i) = NaN;
129     else
130         denom150 = p2 * vel150(i)^2 * area150;
131         cd150(i) = 2 * forcedrag150(i) / denom150;
132         cl150(i) = 2 * forcelift150(i) / denom150;
133         cdCI150(i) = 2 * dragCIforce150(i) / denom150;
134         clCI150(i) = 2 * liftCIforce150(i) / denom150;
135     end
136
137     if vel165(i) == 0
138         cd165(i) = NaN;
139         cl165(i) = NaN;
140         cdCI165(i) = NaN;
141         clCI165(i) = NaN;
142     else
143         denom165 = p1 * vel165(i)^2 * area165;
144         cd165(i) = 2 * forcedrag165(i) / denom165;
145         cl165(i) = 2 * forcelift165(i) / denom165;
146         cdCI165(i) = 2 * dragCIforce165(i) / denom165;
147         clCI165(i) = 2 * liftCIforce165(i) / denom165;
148     end
149
150     if vel180(i) == 0
151         cd180(i) = NaN;
152         cl180(i) = NaN;
153         cdCI180(i) = NaN;
154         clCI180(i) = NaN;
155     else
156         denom180 = p1 * vel180(i)^2 * area180;
157         cd180(i) = 2 * forcedrag180(i) / denom180;
158         cl180(i) = 2 * forcelift180(i) / denom180;
159         cdCI180(i) = 2 * dragCIforce180(i) / denom180;
160         clCI180(i) = 2 * liftCIforce180(i) / denom180;
161     end
162 end
163
164 L180 = 0.1743;
165 L165 = 0.17096;
166 L150 = 0.16245;
167
168 Re150 = vel150 * L150 / (1.55e-5);
169 Re165 = vel165 * L165 / (1.54e-5);
170 Re180 = vel180 * L180 / (1.54e-5);
171

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172 % Drag coefficient plot with 95% CI error bars
173 figure;
174 errorbar(Re150, cd150, cdCI150, '-o', 'LineWidth', 1.2, 'CapSize', 6);
175 hold on;
176 errorbar(Re165, cd165, cdCI165, '-o', 'LineWidth', 1.2, 'CapSize', 6);
177 errorbar(Re180, cd180, cdCI180, '-o', 'LineWidth', 1.2, 'CapSize', 6);
178 legend('150 Degree Bend', '165 Degree Bend', '180 Degree Bend', 'Location', 'best');
179 xlabel('Reynolds Number');
180 ylabel('Drag Coefficient');
181 title('Drag Coefficient vs Reynolds Number for Varying Hip Bend Angles');
182 grid on;
183
184 % Lift coefficient plot with 95% CI error bars
185 figure;
186 errorbar(Re150, cl150, clCI150, '-o', 'LineWidth', 1.2, 'CapSize', 6);
187 hold on;
188 errorbar(Re165, cl165, clCI165, '-o', 'LineWidth', 1.2, 'CapSize', 6);
189 errorbar(Re180, cl180, clCI180, '-o', 'LineWidth', 1.2, 'CapSize', 6);
190 legend('150 Degree Bend', '165 Degree Bend', '180 Degree Bend', 'Location', 'best');
191 xlabel('Reynolds Number');
192 ylabel('Lift Coefficient');
193 title('Lift Coefficient vs Reynolds Number for Varying Hip Bend Angles');
194 grid on;
195
196
197 %% Human-scale wind force prediction
198
199 scale = 10;
200
201 Lhuman150 = scale * L150;
202 Lhuman165 = scale * L165;
203 Lhuman180 = scale * L180;
204
205 Ahuman150 = scale^2 * area150;
206 Ahuman165 = scale^2 * area165;
207 Ahuman180 = scale^2 * area180;
208
209 rhoHuman = 1.184;
210 nuHuman = 1.54e-5;
211
212 Vhuman_mph = linspace(0, 100, 200);
213 Vhuman = Vhuman_mph * 0.44704;
214
215 % Drag fits use original saturating fit
216 dragFit150 = fitSaturatingCoeff(Re150, cd150);
217 dragFit165 = fitSaturatingCoeff(Re165, cd165);
218 dragFit180 = fitSaturatingCoeff(Re180, cd180);
219
220 % Lift fits
221 % Only 165 degree bend uses positive large-Re fit
222 % 180 degree bend uses constant CL to smooth lift force
223 liftFit150 = fitSaturatingCoeff(Re150, cl150);
224 liftFit165 = fitPositiveSaturatingCoeff(Re165, cl165);
225 CL180_const = mean(cl180(isfinite(cl180)));
226
227 ReHuman150 = Vhuman * Lhuman150 / nuHuman;
228 ReHuman165 = Vhuman * Lhuman165 / nuHuman;
229 ReHuman180 = Vhuman * Lhuman180 / nuHuman;
230
231 CdHuman150 = evalSaturatingCoeff(dragFit150, ReHuman150);
232 CdHuman165 = evalSaturatingCoeff(dragFit165, ReHuman165);

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233 CdHuman180 = evalSaturatingCoeff(dragFit180, ReHuman180);
234
235 ClHuman150 = evalSaturatingCoeff(liftFit150, ReHuman150);
236 ClHuman165 = evalSaturatingCoeff(liftFit165, ReHuman165);
237 ClHuman180 = CL180_const * ones(size(Vhuman));
238
239 DragHuman150 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman150 .* CdHuman150;
240 DragHuman165 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman165 .* CdHuman165;
241 DragHuman180 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman180 .* CdHuman180;
242
243 LiftHuman150 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman150 .* ClHuman150;
244 LiftHuman165 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman165 .* ClHuman165;
245 LiftHuman180 = 0.5 * rhoHuman .* Vhuman.^2 .* Ahuman180 .* ClHuman180;
246
247 figure;
248 plot(Vhuman_mph, DragHuman150, 'LineWidth', 1.5);
249 hold on;
250 plot(Vhuman_mph, DragHuman165, 'LineWidth', 1.5);
251 plot(Vhuman_mph, DragHuman180, 'LineWidth', 1.5);
252 xlabel('Wind Speed (mph)');
253 ylabel('Drag Force on Human (N)');
254 title('Predicted Human Drag Force vs Wind Speed');
255 legend('150 Degree Bend', '165 Degree Bend', '180 Degree Bend', 'Location', 'best');
256 grid on;
257
258 figure;
259 plot(Vhuman_mph, LiftHuman150, 'LineWidth', 1.5);
260 hold on;
261 plot(Vhuman_mph, LiftHuman165, 'LineWidth', 1.5);
262 plot(Vhuman_mph, LiftHuman180, 'LineWidth', 1.5);
263 xlabel('Wind Speed (mph)');
264 ylabel('Lift Force on Human (N)');
265 title('Predicted Human Lift Force vs Wind Speed');
266 legend('150 Degree Bend', '165 Degree Bend', '180 Degree Bend - Constant CL', 'Location',
'best');
267 grid on;
268
269 figure;
270 plot(Vhuman_mph, ClHuman150, 'LineWidth', 1.5);
271 hold on;
272 plot(Vhuman_mph, ClHuman165, 'LineWidth', 1.5);
273 plot(Vhuman_mph, ClHuman180, 'LineWidth', 1.5);
274 xlabel('Wind Speed (mph)');
275 ylabel('Extrapolated Lift Coefficient');
276 title('Human-Scale Lift Coefficient vs Wind Speed');
277 legend('150 Degree Bend', '165 Degree Bend - Positive Fit', '180 Degree Bend - Constant CL',
'Location', 'best');
278 grid on;
279
280 fprintf('\nDrag coefficient equations: C_D = C_inf + A*exp(-Re/Re0)\n');
281 fprintf('150 deg: C_D = %.4f + %.4f*exp(-Re/Re0)\n', dragFit150(1), dragFit150(2),
dragFit150(3));
282 fprintf('165 deg: C_D = %.4f + %.4f*exp(-Re/Re0)\n', dragFit165(1), dragFit165(2),
dragFit165(3));
283 fprintf('180 deg: C_D = %.4f + %.4f*exp(-Re/Re0)\n', dragFit180(1), dragFit180(2),
dragFit180(3));
284
285 fprintf('\nLift coefficient equations:\n');
286 fprintf('150 deg: C_L = %.4f + %.4f*exp(-Re/Re0)\n', liftFit150(1), liftFit150(2),
liftFit150(3));

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287 fprintf('165 deg: C_L = %.4f + %.4f*exp(-Re/%.4e)  positive large-Re fit\n', liftFit165(1),
liftFit165(2), liftFit165(3));
288 fprintf('180 deg: C_L = %.4f  constant smooth fit\n', CL180_const);
289
290 fprintf('\nHuman-scale CL ranges:\n');
291 fprintf('150 deg CL: min = %.4f, max = %.4f\n', min(CLHuman150), max(CLHuman150));
292 fprintf('165 deg CL: min = %.4f, max = %.4f\n', min(CLHuman165), max(CLHuman165));
293 fprintf('180 deg CL: min = %.4f, max = %.4f\n', min(CLHuman180), max(CLHuman180));
294
295
296 %% Unsafe wind speed check using tractive limit
297
298 mu = 0.9;
299
300 weightMan_lb = 199;
301 weightWoman_lb = 171;
302
303 lb_to_N = 4.44822;
304
305 Wman = weightMan_lb * lb_to_N;
306 Wwoman = weightWoman_lb * lb_to_N;
307
308 tractiveMan150 = mu * (Wman - LiftHuman150);
309 tractiveMan165 = mu * (Wman - LiftHuman165);
310 tractiveMan180 = mu * (Wman - LiftHuman180);
311
312 tractiveWoman150 = mu * (Wwoman - LiftHuman150);
313 tractiveWoman165 = mu * (Wwoman - LiftHuman165);
314 tractiveWoman180 = mu * (Wwoman - LiftHuman180);
315
316 unsafeMan150 = DragHuman150 > tractiveMan150;
317 unsafeMan165 = DragHuman165 > tractiveMan165;
318 unsafeMan180 = DragHuman180 > tractiveMan180;
319
320 unsafeWoman150 = DragHuman150 > tractiveWoman150;
321 unsafeWoman165 = DragHuman165 > tractiveWoman165;
322 unsafeWoman180 = DragHuman180 > tractiveWoman180;
323
324 unsafeWindMan150 = findFirstUnsafeSpeed(Vhuman_mph, unsafeMan150);
325 unsafeWindMan165 = findFirstUnsafeSpeed(Vhuman_mph, unsafeMan165);
326 unsafeWindMan180 = findFirstUnsafeSpeed(Vhuman_mph, unsafeMan180);
327
328 unsafeWindWoman150 = findFirstUnsafeSpeed(Vhuman_mph, unsafeWoman150);
329 unsafeWindWoman165 = findFirstUnsafeSpeed(Vhuman_mph, unsafeWoman165);
330 unsafeWindWoman180 = findFirstUnsafeSpeed(Vhuman_mph, unsafeWoman180);
331
332 fprintf('\nUnsafe wind speeds using tractive limit:\n');
333 fprintf('Unsafe when Drag Force > mu*(Weight - Lift Force)\n');
334 fprintf('mu = %.2f\n\n', mu);
335
336 fprintf('Man, %.0f lb:\n', weightMan_lb);
337 fprintf('150 deg: %.2f mph\n', unsafeWindMan150);
338 fprintf('165 deg: %.2f mph\n', unsafeWindMan165);
339 fprintf('180 deg: %.2f mph\n\n', unsafeWindMan180);
340
341 fprintf('Woman, %.0f lb:\n', weightWoman_lb);
342 fprintf('150 deg: %.2f mph\n', unsafeWindWoman150);
343 fprintf('165 deg: %.2f mph\n', unsafeWindWoman165);
344 fprintf('180 deg: %.2f mph\n', unsafeWindWoman180);
345
346 figure;

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347 plot(Vhuman_mph, DragHuman150, 'LineWidth', 1.5);
348 hold on;
349 plot(Vhuman_mph, tractiveMan150, '--', 'LineWidth', 1.5);
350 plot(Vhuman_mph, tractiveWoman150, '--', 'LineWidth', 1.5);
351 xlabel('Wind Speed (mph)');
352 ylabel('Force (N)');
353 title('Tractive Limit Check, 150 Degree Bend');
354 legend('Drag Force', 'Man Tractive Limit', 'Woman Tractive Limit', 'Location', 'best');
355 grid on;
356
357 figure;
358 plot(Vhuman_mph, DragHuman165, 'LineWidth', 1.5);
359 hold on;
360 plot(Vhuman_mph, tractiveMan165, '--', 'LineWidth', 1.5);
361 plot(Vhuman_mph, tractiveWoman165, '--', 'LineWidth', 1.5);
362 xlabel('Wind Speed (mph)');
363 ylabel('Force (N)');
364 title('Tractive Limit Check, 165 Degree Bend');
365 legend('Drag Force', 'Man Tractive Limit', 'Woman Tractive Limit', 'Location', 'best');
366 grid on;
367
368 figure;
369 plot(Vhuman_mph, DragHuman180, 'LineWidth', 1.5);
370 hold on;
371 plot(Vhuman_mph, tractiveMan180, '--', 'LineWidth', 1.5);
372 plot(Vhuman_mph, tractiveWoman180, '--', 'LineWidth', 1.5);
373 xlabel('Wind Speed (mph)');
374 ylabel('Force (N)');
375 title('Tractive Limit Check, 180 Degree Bend');
376 legend('Drag Force', 'Man Tractive Limit', 'Woman Tractive Limit', 'Location', 'best');
377 grid on;
378
379
380 %% Local functions
381
382 function params = fitSaturatingCoeff(Re, C)
383     valid = isfinite(Re) & isfinite(C) & Re > 0;
384     Re = Re(valid);
385     C = C(valid);
386
387     Cinf0 = mean(C(end-2:end));
388     A0 = C(1) - Cinf0;
389     Re00 = mean(Re);
390
391     p0 = [Cinf0, A0, Re00];
392
393     obj = @(p) sum((C - (p(1) + p(2)*exp(-Re./abs(p(3)))).^2);
394
395     params = fminsearch(obj, p0);
396     params(3) = abs(params(3));
397 end
398
399 function params = fitPositiveSaturatingCoeff(Re, C)
400     valid = isfinite(Re) & isfinite(C) & Re > 0 & C > 0;
401     Re = Re(valid);
402     C = C(valid);
403
404     Cinf0 = max(mean(C(end-2:end)), 0.001);
405     A0 = C(1) - Cinf0;
406     Re00 = mean(Re);
407

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408     p0 = [log(Cinf0), A0, log(Re00)];
409
410     obj = @(p) sum((C - (exp(p(1)) + p(2)*exp(-Re./exp(p(3)))).^2);
411
412     p = fminsearch(obj, p0);
413
414     Cinf = exp(p(1));
415     A     = p(2);
416     Re0   = exp(p(3));
417
418     params = [Cinf, A, Re0];
419 end
420
421 function C = evalSaturatingCoeff(params, Re)
422     Cinf = params(1);
423     A     = params(2);
424     Re0   = params(3);
425
426     C = Cinf + A .* exp(-Re ./ Re0);
427 end
428
429 function unsafeSpeed = findFirstUnsafeSpeed(V, unsafeArray)
430     idx = find(unsafeArray, 1, 'first');
431
432     if isempty(idx)
433         unsafeSpeed = NaN;
434     else
435         unsafeSpeed = V(idx);
436     end
437 end

```